

Confluent Kafka Isotope

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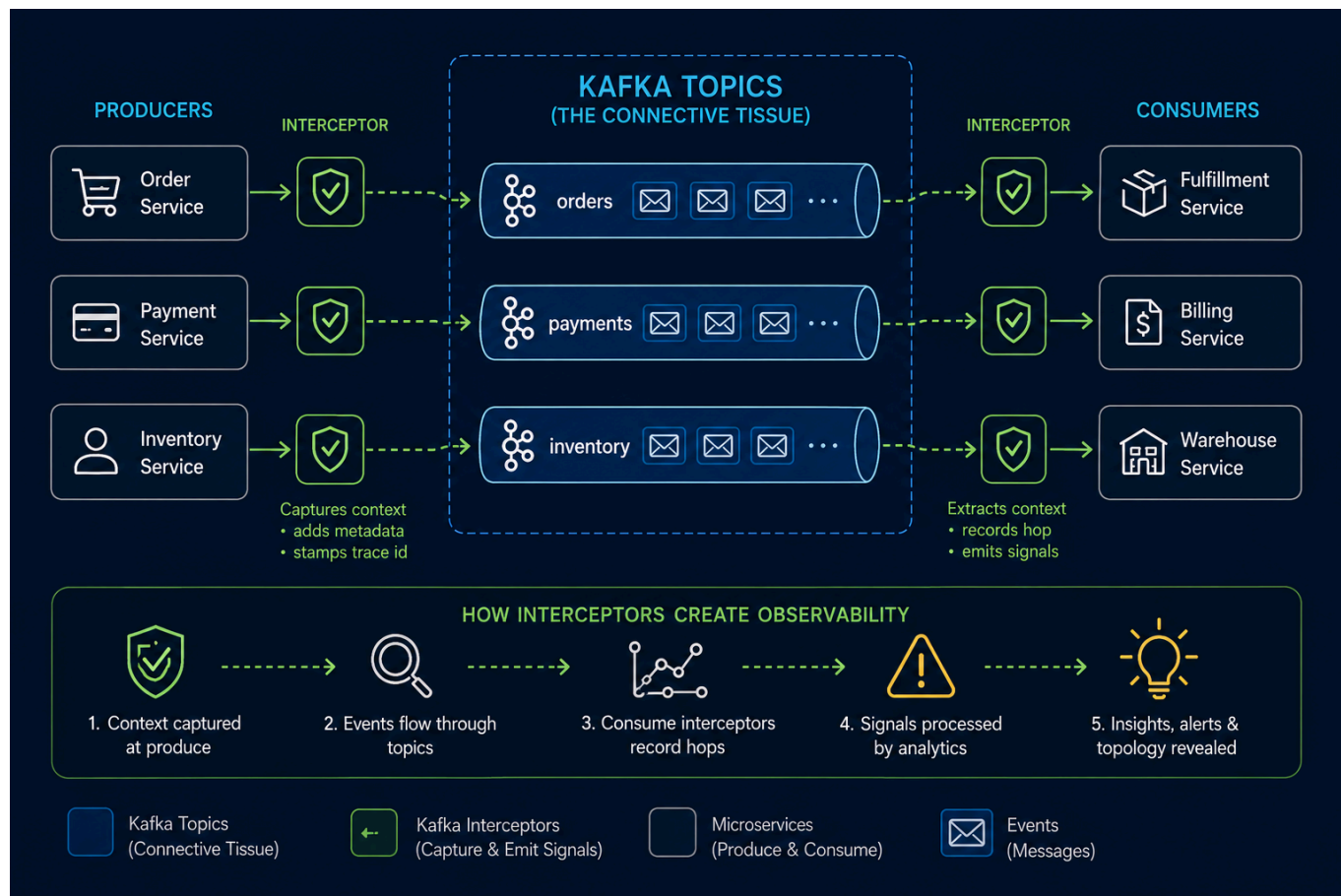
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Purpose

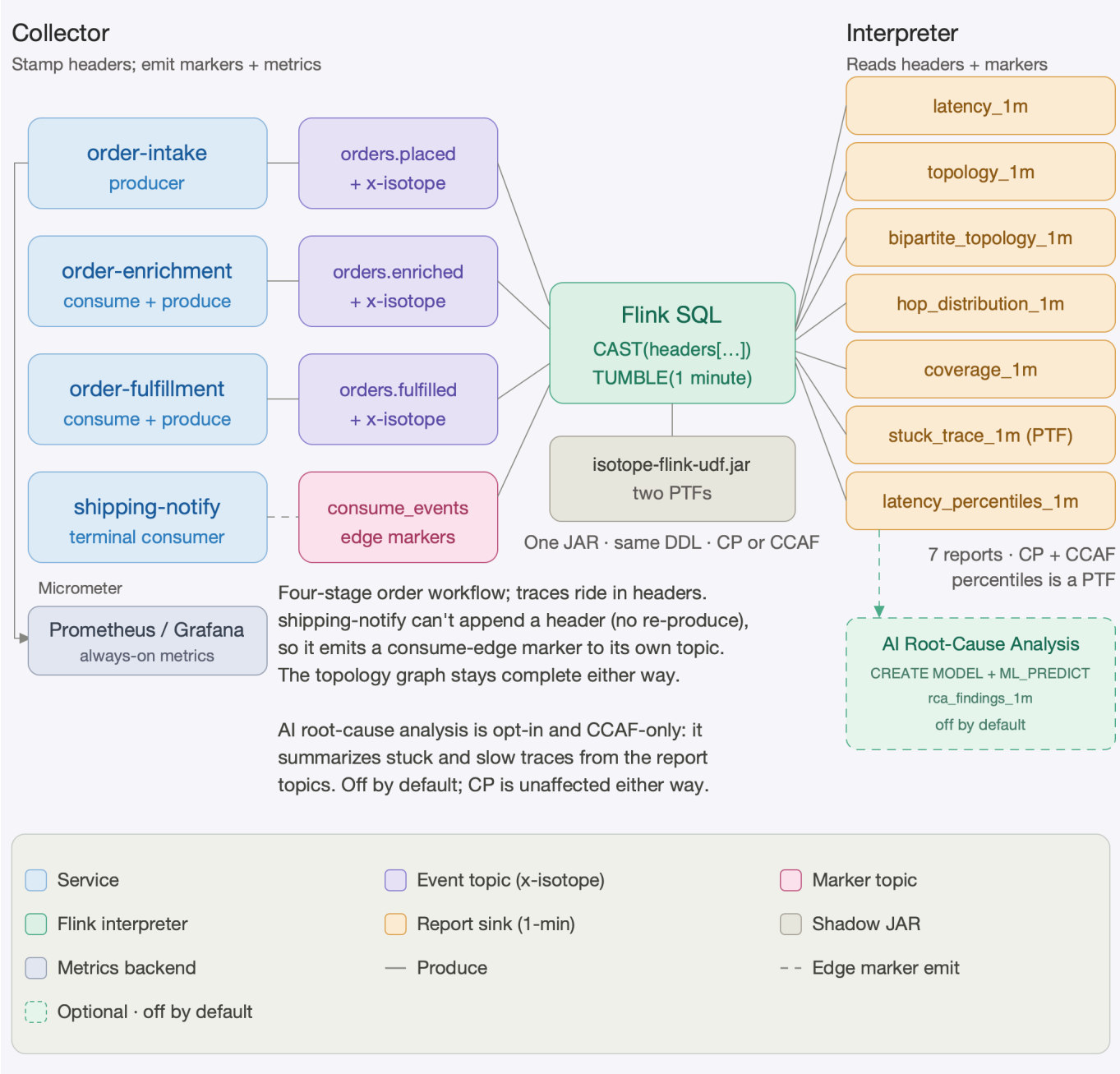
`confluent-kafka-isotope` is a reference implementation of an **e-commerce order pipeline that uses Kafka Interceptors, Prometheus, and Apache Flink to capture, analyze, and report end-to-end event-tracing data in both batch and near real time.**

Much like isotopes used to trace molecules through a biochemical pathway, each event carries lightweight metadata that allows it to be followed as it travels through Kafka topics and distributed microservices.

Kafka topics become the **connective tissue between services**, while **Kafka Interceptors quietly transform** the event pipeline itself into an **observable distributed system**. (As depicted in the diagram below.)



This project demonstrates how **Kafka Interceptors become collectors** — inserting the isotope into record headers in place and, at terminal consumers, emitting consume-edge markers — while **Flink SQL serves as the interpreter**, reading those headers directly to produce seven 1-minute reports from a single JAR on both Confluent Platform and Confluent Cloud for Apache Flink (CCAF). Optionally, the producer interceptor can **emit Micrometer metrics to Prometheus as an always-on aggregate layer** alongside the per-trace Flink reports. (As depicted in the diagram below.)



With this approach, developers gain **end-to-end observability** into the flow of events through the Kafka-based microservices architecture, enabling both **real-time monitoring** and **post-hoc analysis** of event traces.

This end-to-end observability of the isotope tracing pipeline creates a **ladder of insights**, allowing developers to trace each event’s journey, identify bottlenecks, and improve the performance and reliability of the entire system. The ladder is organized into **five tiers that answer 30 questions**. (As depicted in the diagram below.)



- ▶ Easy — single-record, single-trace
- ▶ Easy to Medium — single per-minute aggregates
- ▶ Medium — cross-window deltas, anomalies, multi-report joins
- ▶ Hard — tail behavior, drift detection, correlated analysis
- ▶ Harder — forensic replay, compliance, cross-system joins

1.0 How an Isotope Traverses an Event Pipeline

An **Isotope** is a *lightweight tracing artifact attached to Kafka record headers*. Like a biochemical isotope used to trace molecules through a metabolic pathway, it allows the journey of a record through an event-driven architecture to be observed and analyzed.

This project models a Kafka pipeline as a **bipartite graph**: *services occupy one vertex set, topics the other, and every produce and consume operation forms an edge between them*. The resulting graph provides a unified view of the complete event topology, from producers to intermediate processing services to terminal consumers.

A **ProducerInterceptor** stamps each record with an isotope and appends one hop for every `send()` operation, capturing the produce edges. Consumers call `IsotopeContext.recordConsume()` to emit a lightweight marker representing the corresponding consume edges, while services that consume and then produce invoke `IsotopeContext.adoptFromRecord()` so the trace identity persists across every

hop. Apache Flink reconstructs the complete **service** → **topic** → **service** graph from the isotope headers alone, producing seven reports: end-to-end latency, latency percentiles, produce-side topology, the full bipartite topology, hop distribution, per-topic coverage (a trace-loss funnel signal), and stuck-trace detection. The same implementation runs unchanged on both Confluent Platform (self-managed) with Confluent Manager for Apache Flink (CMF) and Confluent Cloud for Apache Flink (CCAF).

Full architectural design of Isotope Tracing — the header layout (`x-isotope` JSON + seven scalar headers with a worked example), how the producer interceptor gets invoked, why the consume side uses explicit calls instead of a `ConsumerInterceptor`, and the bipartite-graph rationale — are documented in [docs/design.md](#).

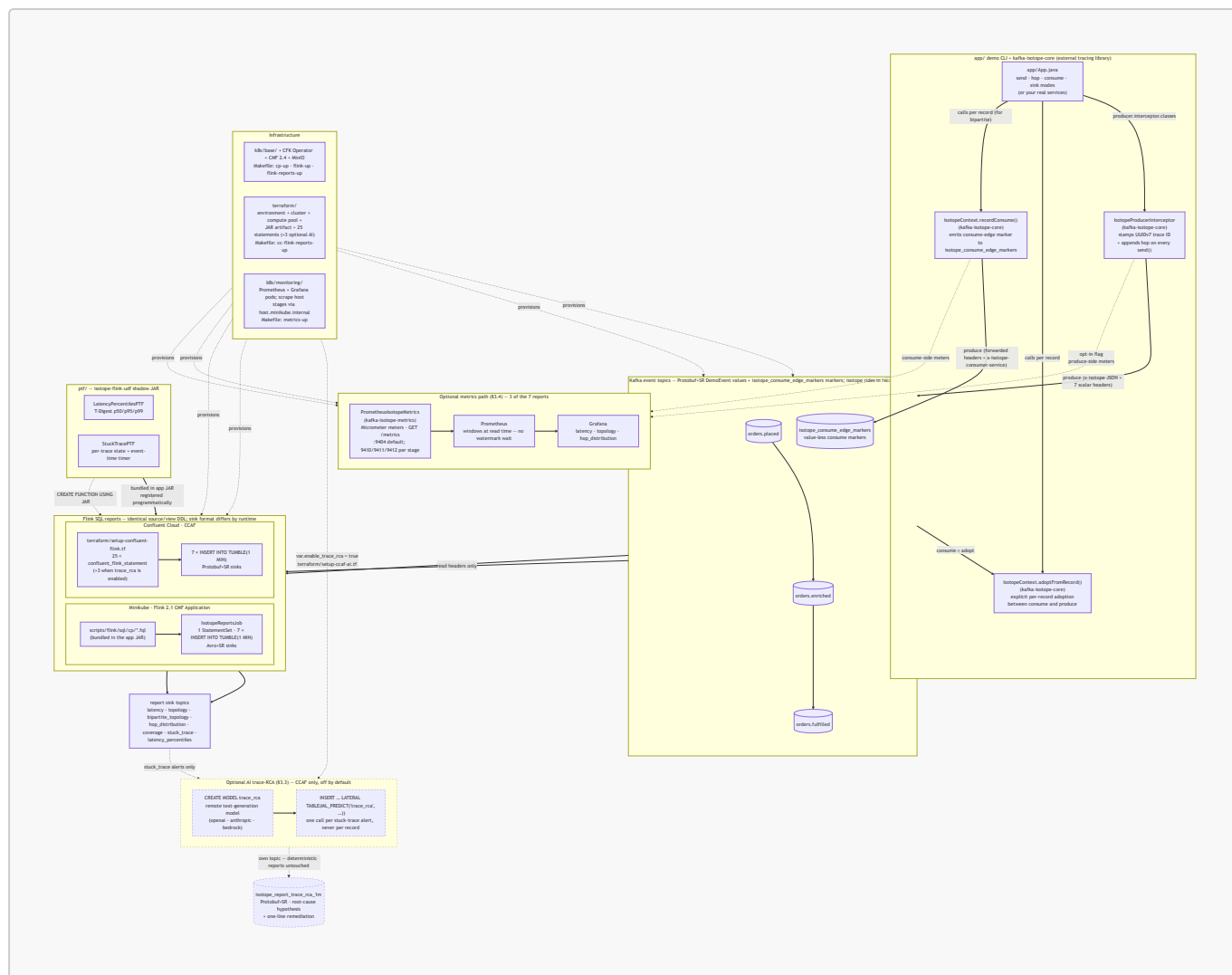
2.0 Architecture

A bird's-eye view of the moving parts. The demo CLI in `app/` consumes the external tracing library (`ai.signalroom:kafka-isotope-core`), which registers a Kafka `ProducerInterceptor` that stamps an isotope into record headers on every `send()`. Consume-then-produce services propagate the inbound trace by explicitly calling `IsotopeContext.adoptFromRecord(record)`. Business events then flow through a three-topic Kafka pipeline, where Flink SQL reads the isotope metadata and emits one-minute aggregate reports.

Both runtimes run the same seven logical reports off the same source and view definitions, though each runtime keeps its own copy of the SQL. **Confluent Platform (CP) + Flink** on Minikube executes the `.fql` files under `scripts/flink/sql/cp/` as a single Flink 2.1 Confluent Manager for Apache Flink (CMF) Application (`IsotopeReportsJob`), while **Confluent Cloud for Apache Flink (CCAF)** applies the same logical SQL as inline `confluent_flink_statement` Terraform resources under `terraform/`. The shadow JAR from `ptf/`—which powers two of the seven reports—runs unchanged on both runtimes: bundled into the CP application JAR and uploaded as a Flink artifact on CCAF.

Alongside that—**additive, opt-in, and disabled by default**—the interceptor can also emit **Micrometer** metrics for **Prometheus**, with **Grafana** providing visualization (§3.4). This path produces the three **stateless** reports (`latency_1m`, `topology_1m`, and `hop_distribution_1m`) without requiring a stream processor. The remaining four reports continue to run in Flink because they depend on per-`trace_id` state or absence-of-event analysis, which falls outside Prometheus's query model.

(Kafka is drawn once below for brevity; each runtime provisions its own Kafka cluster.)



► See Repo Layout

3.0 Getting Started

First, clone the repo to your local machine using the [GitHub CLI](#):

```
gh repo clone j3-signalroom/confluent-kafka-isotope
```

Change to the repo directory:

```
cd /path/to/confluent-kafka-isotope
```

Then decide how you want to run the repo:

3.1 Integration Tests with Confluent Platform on Minikube

```
make install-prereqs # docker, kubectl, minikube, helm, gettext, gradle, openjdk17
```

```
make check-prereqs      # verify they're on PATH
```

Default Minikube sizing (override via env): **MINIKUBE_CPUS=6**, **MINIKUBE_MEM=20480**, **MINIKUBE_DISK=50g**.

Bring up the local Confluent Platform stack and port-forward Kafka + SR:

```
make minikube-start      # one-time
make cp-up               # CFK Operator +
Kafka/SR/Connect/ksqlDB/C3 (~5 min)
make kafka-pf-up         # localhost:30092 → Kafka,
localhost:8081 → Schema Registry
```

Then run the suite:

```
./gradlew :app:integrationTest      #
every IT below
./gradlew :app:integrationTest --tests '*ProducerInterceptorIT'      #
just one
```

Override the endpoints if needed:

```
./gradlew :app:integrationTest \
  -PkafkaBootstrap=localhost:30092 \
  -PschemaRegistryUrl=http://localhost:8081
```

Tear down forwards when done:

```
make kafka-pf-down
```

The integration tests cover:

Test	What it verifies
BrokerSmokeIT	AdminClient can create/list/delete a topic via the NodePort port-forward
ProducerInterceptorIT	A consumer sees the x-isotope JSON header + all 7 scalar reporting headers with the expected origin/hop values, and the Protobuf round-trip preserves DemoEvent.source / payload
ThreeStageHopPropagationIT	order-intake-service → topic-AB → order-enrichment-service → topic-BC → order-fulfillment-service produces a stable trace ID, 2-hop trail in send order,

Test	What it verifies
	and correct scalar headers (origin = <code>order-intake-service</code> , this = <code>order-enrichment-service</code> , hop count = 2) at the terminal; consume-then-produce hops use <code>IsotopeContext.adaptFromRecord</code> to carry the trace forward
<code>BipartiteTopologyIT</code>	The 4-stage <code>order-intake-service</code> → <code>topic-AB</code> → <code>order-enrichment-service</code> → <code>topic-BC</code> → <code>order-fulfillment-service</code> → <code>topic-CD</code> → <code>shipping-notification-service</code> chain emits exactly three consume-edge markers to a per-test markers topic — one per consume edge. Every marker carries the trace ID, forwarded <code>x-isotope-*</code> scalars describing the upstream producer, and the new <code>x-isotope-consumer-service</code> naming the downstream consumer. Asserts the <code>(consumer_service, consumed_topic)</code> set is exactly the three pairs of stages 2-4

3.2 Seven Scalar Headers Flink SQL Reports with Apache Flink on Minikube

Caveat: Minikube is a single-node cluster. It is not a production-like environment, but it is sufficient for local development and testing. The Confluent Platform (CP) + Flink stack runs in Minikube, and you can port-forward Kafka and Schema Registry to your local machine.

To **run**, **test**, and **debug** Apache Flink like a production engineer, this project provides a full Confluent Platform + Flink stack running locally on [Minikube](#) — no cloud required.

You get an environment on your machine, with all the components you'd expect in a real deployment:

- **Confluent Platform** (KRaft mode) via Confluent for Kubernetes (CFK)
- **Apache Flink 2.1.2** via the Confluent Flink Kubernetes Operator 1.140.1
- **Confluent Manager for Apache Flink (CMF) 2.4.0** for Flink environment management

To run this project, you'll need **macOS (with Homebrew)** or **Linux (with apt-get)**. The full stack — **Minikube + Confluent Platform + Flink + CMF** — is resource-intensive and designed to mirror an adequate development environment. Therefore, the following defaults are recommended:

Resource	Default
CPU	6
Memory	20 GB
Disk	50 GB

These settings ensure stable performance across all components. You can tune them as needed, but lower resource levels may cause pod restarts or degraded performance.

Seven reports — five pure Flink SQL plus two JAR-backed PTFs — run as a **single Flink 2.1 CMF Application** (`IsotopeReportsJob`, one StatementSet, seven sinks) submitted through CMF and executed by the Confluent Flink Kubernetes Operator. No raw session cluster is involved;

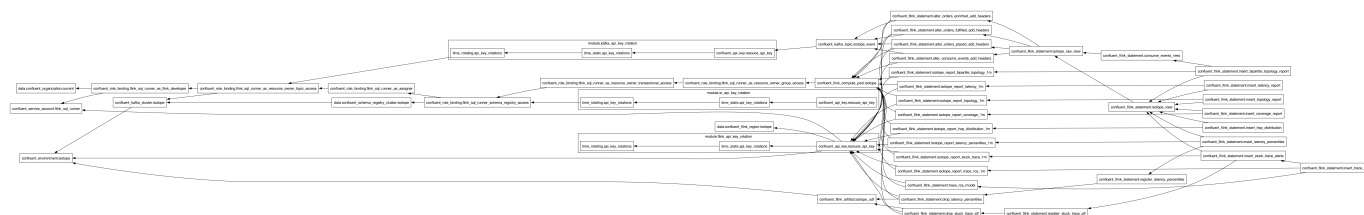
`k8s/base/flink-cluster-deployment.yaml` (`make flink-deploy` / `make flink-sql`) is a separate, optional path for ad-hoc SQL. Confluent Cloud runs the same seven reports from its own copy of the SQL, inlined in Terraform — see §3.3 for that path; this section is the local-Minikube one.

The full bring-up sequence — cluster → Flink → reports → traffic → teardown — is consolidated in [docs/runbook-minikube.md](#). The short version: `make flink-up` then `make flink-reports-up`, then drive traffic across **multiple** 1-minute windows (a single burst sits in one open window forever — the watermark has to cross `window_end` for a tumbling window to emit) and wait ~90s after the last record.

Report sink topics ride **Avro+SR** (`avro-confluent`, auto-registered on first write) so Control Center renders them natively — a deliberate *format-by-domain* split: app events are **Protobuf+SR** (`DemoEvent`), Flink aggregates are **Avro+SR** (cp-flink ships no SR-integrated Protobuf format), and the consume-edge marker topic `isotope_consume_edge_markers` is **null-value / headers-only**. Not a defect — a clean split by domain.

3.3 Seven Scalar Headers Flink SQL Reports with Confluent Cloud for Apache Flink

The Confluent Cloud for Apache Flink (CCAF) parallel of §3.2 CP + Apache Flink on MiniKube, driven by Terraform under [terraform/](#). The Terraform graph below depicts the resources deployed in Confluent Cloud.



`make cc-flink-reports-up` provisions a fresh Confluent Cloud environment containing:

- Kafka cluster
- Compute pool
- Two rotating service-account API key pairs (Kafka + Schema Registry)
- The uploaded PTF JAR
- 25 `confluent_flink_statement` resources (4 `ALTER TABLE` + 3 `VIEW` + 7 sink `CREATE TABLE` + 2 `CREATE FUNCTION` + 7 `INSERT INTO`, plus 2 transient `DROP FUNCTION`)

The full provision → deploy → traffic → teardown sequence is consolidated in [docs/runbook-ccaf.md](#). `make cc-flink-reports-up` (~6–8 min first run; idempotent re-applies), drive traffic with `scripts/cc-app-run.sh place|enrich|fulfill|ship` across **multiple** 1-minute windows (a single burst sits in one open window forever, and you wait ~90s after the last record), then `make cc-flink-reports-down` to `terraform destroy` the whole environment.

Format-by-runtime (not-by-domain). CP's reports land on **Avro+SR** (`'value.format' = 'avro-confluent'` in `scripts/flink/sql/cp/05_report_sinks.fql`). CCAF's reports land on **Protobuf+SR** (`'value.format' = 'proto-registry'` in each sink's `WITH` clause in `terraform/setup-confluent-flink.tf`). The two runtimes' SQL is otherwise unshared: CP's lives hardcoded in `scripts/flink/sql/cp/`, CCAF's lives inline as `confluent_flink_statement` resources in `terraform/setup-confluent-flink.tf`.

3.3.1 Why `latency_percentiles` is a `ProcessTableFunction` (PTF)

Since CCAF does not support [User-Defined AGGregate functions \(UDAGG\)](#), I implemented the percentiles report as a [ProcessTableFunction](#) to keep it portable across runtimes.

[LATENCY_PERCENTILES](#) (class [LatencyPercentilesPTF](#)) performs its own 1-minute tumbling-window aggregation over a T-Digest sketch using per-window state and event-time timers. **A PTF avoids that UDAGG restriction, so it registers and runs on both runtimes, just like [STUCK_TRACE_PTF](#). Both runtimes therefore run the same seven reports:** [latency](#) (avg/min/max), [topology](#) (produce-side), [bipartite_topology](#) (full service↔topic↔service graph), [hop_distribution](#), [coverage](#), [stuck_trace](#), and [latency_percentiles](#) (p50/p95/p99).

3.3.2 [OPTIONAL] Eighth Report — AI Root-Cause Analysis (RCA)

Beyond the seven deterministic reports, [terraform/setup-ccaf-ai.tf](#) wires an **eighth, AI-generated report** that turns each stuck-trace *alert* into a natural-language root-cause hypothesis plus a one-line remediation. It adds three [confluent_flink_statement](#) resources — [CREATE MODEL trace_rca](#) (a remote text-generation model), a [proto-registry](#) sink [isotope_report_trace_rca_1m](#), and an [INSERT ... SELECT ... LATERAL TABLE\(ML_PREDICT\('trace_rca', ...\)\)](#) — all **gated on [var.enable_trace_rca](#) (default [false](#))**, so a normal [make cc-flink-reports-up](#) is completely unaffected. The model is invoked once per *alert* (the low-volume 1-minute stuck-trace report topic), never per record, and its output lands on its own topic — it never overwrites the deterministic reports. The defaults target OpenAI ([gpt-4o](#)). The standalone SQL PoC — a [CREATE MODEL + ML_PREDICT](#) walkthrough with provider notes and alternative options — is in [scripts/flink/sql/ccaf-ai/trace_rca.fql](#).

3.4 [OPTIONAL] Prometheus Metrics Reporting with Grafana Visualization

Three of the seven reports — [latency_1m](#), [topology_1m](#), [hop_distribution_1m](#) — are pure stateless scalar aggregation over bounded-cardinality dimensions (service / topic / hop_count, never [trace_id](#)). Those don't need a stream processor: the [producer interceptor](#) already has every value in scope on each [send\(\)](#), so it emits them as **Micrometer** meters ([PrometheusIsotopeMetrics.java](#)) and lets **Prometheus** window them at read time, with **Grafana** on top. The other four ([latency_percentiles](#), [coverage](#), [bipartite_topology](#), [stuck_trace](#)) are per-[trace_id](#) stateful or absence-of-event problems Prometheus can't express, so they **stay in Flink**. This path is **additive and opt-in** — a 3-Micrometer / 4-Flink split.

Full details — the meter/PromQL reference, the produce- and consume-side signals, the two deliberate gaps ([distinct_traces](#), windowed [min](#)), and why percentiles stays a PTF — are in [docs/metrics.md](#). The one-command Prometheus + Grafana showcase has its own runbook: [k8s/monitoring/README.md](#) ([make metrics-up](#)).

Resources

- [Medium Article: Kafka's quiet observability superpower — Kafka Interceptors](#)